

Lab Questions (2 Hours)

1. Accessing and Processing Data from APIs (REST, SOAP)

- 1 Write a Python program to consume a REST API and display JSON data.
- 2 Write a program to fetch data from a REST API and count total records.
- 3 Write a Python program to extract specific fields from a REST API response.
- 4 Write a program to access a SOAP web service using Python.
- 5 Write a Python program to perform an arithmetic operation using a SOAP API.
- 6 Write a program to handle HTTP status codes while accessing an API.

2. Web Scraping using Requests and BeautifulSoup

- 7 Write a Python program to fetch HTML content from a web page.
- 8 Write a program to extract the title of a web page using BeautifulSoup.
- 9 Write a Python program to extract all hyperlinks from a webpage.
- 10 Write a program to scrape tabular data from a website.
- 11 Write a Python program to extract text from specific HTML tags.
- 12 Write a program to store scraped data into a file.

3. Handling Large Datasets (Chunking and Lazy Evaluation)

- 13 Write a Python program to read a large CSV file using chunking.
- 14 Write a program to process large datasets without loading them fully into memory.
- 15 Write a Python program to calculate total rows using chunk-based reading.
- 16 Write a program to demonstrate lazy evaluation using generators.
- 17 Write a Python program to read a large text file using generators.
- 18 Write a program to compare normal file reading and lazy evaluation.